



UNIVERSITI TEKNOLOGI MALAYSIA

TASK 2

NETWORK COMMUNICATION

NO	NAME	MATRIC NO
1	MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	A23CS0250
2	AHMAD ADIB ZIKRI BIN A.MAZLAM	A23CS0205
3	SYARIFAH DANIA BINTI SYED ABU BAKAR	A23CS0183
4	YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201

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1.0 Introduction

The Faculty of Computing is preparing to develop a cutting-edge network plan for its new two-storey building, designed to support the institution's vision for the next generation of technological education. With the expected increase in student and staff numbers by 15% over the coming four years, this expansion project is essential for accommodating both current and future needs. The facility will feature modern labs, collaborative spaces, and specialized learning areas such as a Cisco Network Lab and an embedded IoT lab, all of which require a high-performance, reliable, and scalable network infrastructure.

This task focuses on gathering the necessary requirements and information through research, interviews, and analysis to ensure the development of an efficient and sustainable network plan. The network must support high-speed internet connectivity, seamless virtual collaboration, and secure access to specialized resources. By thoroughly analyzing the needs of the faculty, this report aims to assess the feasibility of the proposed network plan and offer insights into how the infrastructure can support both the immediate and long-term educational goals of the Faculty of Computing

2.0 Question

1. Are there specific software and hardware requirements for each lab?

Yes, each lab has unique needs. The general-purpose labs require office and basic programming software. The embedded lab requires circuit simulation tools, embedded systems kits, and IoT equipment. The Cisco lab needs networking hardware such as routers, switches, and network simulation software. The video conferencing room should have high-quality cameras, microphones, and conferencing software, while the hybrid classroom needs an AV system and tools for remote learning integration.

Reference: Lim, H., Yu, H., & Suh, T. (2009). Using virtual platform in embedded system education. *Computer Applications in Engineering Education*, 20(2), 346–355. <https://doi.org/10.1002/cae.20401>

2. How to ensure that the coverage of wireless networks is able to support the area of the faculty?

We assess the performance of the existing wireless network; first, we identify the locations of most congested access points and secondly we identify problem areas with poor or no signal coverage throughout the campus. To do this, we first generate wireless signal coverage maps in selected locations using wireless measurement tools and, separately, study network data collected from the Wi-Fi management platform. We then identify the busiest campus access points by looking at the number of connected clients to each access point and list the problem areas having wireless signal strength below a reference level. Finally, we propose suitable solutions to the problems and present coverage predictions of the wireless network if the proposed solution were implemented.

Reference: F. B. K. Easha, R. Abbas, and M. Daley, "Campus Wi-Fi coverage mapping and analysis," School of Engineering (Telecommunications), Macquarie University, New South Wales, Australia

3. .What monitoring, management, and reporting tools are needed for network maintenance?

The main goals of network monitoring tools are account, fault, and performance monitoring. They are also employed to analyze parts like email servers and programs. The best network monitoring tools are SolarWinds Network Performance Monitor (NPM). It monitors network problems and sends helpful alerts to the respective teams. To assist users in understanding important network paths, the application offers more than 100 pre-built reports. To quickly track and identify network problems, NPM

displays all of the data pertaining to devices, networks, vendors, and applications in a single-page path analysis. With the automation and customization capabilities of NPM, administrators may additionally plan network performance and availability reports. Users can create network visualisation maps automatically with the PerfStack™ capability. All that we can do is drag and drop the network performance data from several sources onto a single chart. For improved analysis, it also provides a broad overview of network resources. Additionally, NPM enables users to create wireless heatmaps by conducting hop-by-hop network analysis to distant services.

Reference : Network Monitoring: Protocols, Best Practices, and Tools | Tek-Tools

4. Does it necessary to have a server room? If yes, what are the required equipments?

Yes, A server room is a special location where companies keep and maintain computer servers; it's the core location that protects the entire IT setup. The server room can help provide a centralized place for network resources, safeguard critical data from physical harm, and securely store network equipment. These are the basic components of a server room. First, cable management equipment, In order to enable clean cable management, patch panels are necessary. There are various varieties of them, including as feed-through, punch-down, and blank patch panels. To keep servers and other important equipment organized and physically protected, a server room often needs one or more server cabinets or racks. Open-frame, selfcontained, wall-mounted, colocation, and server rack cabinets are among the several varieties of server racks. Server racks guarantee effective cable management and airflow in addition to optimizing server room space. Next, is Routers and switches are essential pieces of network equipment that transfer data between users, servers, and other network-connected devices. Storage devices, firewalls, and other equipment are also essential for maintaining consistent communication, preventing data loss, and ensuring security in server rooms. Lastly, Network Fiber optic cables may be used in server room cabling. Although fiber optic cables might be fragile, they allow interference-free long-distance data transfer. Fiber connectors are used to terminate fiber optic lines. In server rooms, a lot of cables are needed to establish reliable physical connections between equipment.

Reference: Fatima. (2024, October 29). Ultimate Guide To Server Room Design And Equip*RedSwitches*.

<https://www.redswitches.com/blog/server-room-design-and-equipment/>

5. What are the anticipated peak usage hours in the facility, and how will this affect network performance?

Anticipated peak usage hours in the facility are from 9 AM to 4 PM on weekdays, coinciding with high-activity periods such as classes, lab sessions, and meetings, leading to full occupancy in areas like labs, classrooms, and the café. This intense demand necessitates efficient load balancing, traffic management, and adequate bandwidth allocation to maintain optimal network performance and prevent bottlenecks. Strategies such as implementing Quality of Service (QoS) policies, continuous network monitoring, and investing in scalable infrastructure are crucial to ensure a robust and reliable network during these peak times (Ogletree, 2019; White, 2021).

Reference : Managing and Optimizing Networks" by John W. L. Ogletree, 2019
Network Performance and Optimization" by Jane White, 2021

6. What are the specific security requirements for the network, especially in the labs and administrative areas?

Security will be crucial, especially in the labs and administrative offices, where sensitive research data and student information are stored and processed. The network must have secure access controls, such as role-based authentication and VLAN segmentation to separate guest traffic from critical areas. Encryption protocols and regular vulnerability assessments will also be necessary to meet institutional security standards. Additionally, implementing a Zero Trust Network Access (ZTNA) framework can further enhance security by ensuring that every access request is authenticated, authorized, and encrypted, regardless of the user's location. Regular security audits and compliance checks should be conducted to identify and mitigate potential vulnerabilities.

Chandramouli, R. (2022). *Guide to a secure enterprise network landscape*.
<https://doi.org/10.6028/nist.sp.800-215>

7. Where should network switches and routers be positioned within the building to reduce cable lengths and optimize network performance in each lab and the Cafe?

Install Wi-Fi access points that connect to nearby switches, ensuring strong wireless coverage and easy expansion if additional APs are needed in the future. These APs should link back to the network room switch for centralized management and bandwidth allocation. AP should be installed throughout the floor in key locations, particularly in high-traffic areas like each lab and the Cafe, to ensure robust wireless connectivity. Each AP should connect to nearby switches, which will minimize the length of cabling required and reduce signal loss. These switches, in turn, should link back to a central network room switch that manages and allocates bandwidth.

What Is a Wireless Network? - Wired vs Wireless. (2024, October 29). Cisco.
<https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/wireless-network.html>

8. How will network usage be monitored and managed?

Network management software will monitor traffic and detect connectivity issues, especially in the Cisco lab, video conferencing room, and hybrid classroom. This software will help manage bandwidth and ensure resource availability, especially during high-demand periods.

9. What are the specific bandwidth requirements for different areas (labs, student lounge, café)?

Bandwidth requirements for different areas:

- * General Purpose Labs: 100 Mbps per user for data-intensive tasks like simulations.
- * Hybrid Classrooms: 50 Mbps per user to support video conferencing and collaboration.
- * Café and Student Lounge: 10 Mbps per user, catering to leisure activities like browsing and social media.
- * An overall bandwidth of 1–2 Gbps is projected for peak times to support the entire facility.

Reference :

<https://www.sciencedirect.com/topics/computer-science/bandwidth-requirement>

10. What kind of video conferencing or AV systems are planned in rooms labelled for such use (e.g., hybrid classrooms, meeting rooms)?

Hybrid classrooms and meeting rooms will feature high-definition cameras, microphones, and large display screens to support virtual and hybrid learning environments. Dedicated network connections with a minimum bandwidth of 200 Mbps will ensure high-quality, uninterrupted audio and video streaming for meetings and online classes. Additionally, these rooms will be equipped with interactive whiteboards, smart cameras with auto-tracking features, and high-quality sound systems to enhance engagement and collaboration. The use of Audio-Visual over Internet Protocol (AVoIP) technology will provide flexibility and scalability in system design, ensuring seamless distribution of audio and video signals across the network.

Reference :

Israk, W. (2023, July 9). Hybrid Classroom Solutions: Enhancing Remote Learning.

Israk Solutions Sdn. Bhd. <https://www.israk.my/hybrid-classroom-enhancing-learning/>

Av, S. C. (2023, June 2). A Comprehensive Guide to Flawless Audio Visual System Planning - 21st Century AV. *21st Century AV*.
<https://21stcenturyav.com/audio-visual-system-planning/>

3.0 Feasibility

3.1 Economic feasibility

Since we know which access points are the busiest and where coverage is lacking, it allows us to focus investment on areas that will yield the highest impact. This targeted approach ensures that funds are spent in locations where improvements will benefit the most users, thus maximizing the return on investment (ROI). By leveraging wireless measurement tools and existing network data, we can minimize the need for expensive site surveys or equipment replacements, helping control costs. This approach ensures that the assessment phase itself remains cost-effective, which is crucial for economic feasibility.

Furthermore, improved wireless coverage can lead to fewer help desk calls. When network coverage is consistent and reliable, users experience fewer connectivity issues. This reduces the number of calls to the IT help desk for network-related problems, which, in turn, saves time and resources that would otherwise be spent troubleshooting and assisting users. Fewer help desk calls also means less downtime for students, staff, and faculty, which can improve productivity. Plus, it reduced the need for temporary fixes. IT staff often have to implement short-term fixes, like installing additional temporary equipment or adjusting existing setups. These quick fixes require time and resources but don't offer a lasting solution. And possibly lower energy costs if the solution includes energy-efficient access points. These savings contribute to economic feasibility by offsetting the initial costs over time.

3.2 Operational feasibility

The facility is well-prepared to handle operational demands with the right mix of infrastructure, technology, and resources. Each lab is designed to meet specific needs: the Cisco lab is equipped with networking devices, the Embedded lab with circuit kits, and hybrid classrooms with advanced AV systems for both in-person and virtual learning. With strategic Wi-Fi placement and switches set up throughout the building, the network can easily support heavy usage during peak hours, like class times, thanks to a planned 1–2 Gbps of bandwidth. Security measures, such as VLAN segmentation and a Zero Trust Network Access framework, will protect sensitive data, while regular security checks will help maintain data privacy and compliance.

Proactive monitoring with tools like SolarWinds NPM will keep an eye on network health in high-demand areas like labs and video conferencing rooms, helping to identify and fix issues before they affect users. A dedicated server room will keep network resources organised and protected, with fibre optics for efficient data transfer. Although this setup requires a reasonable budget for initial installation and ongoing maintenance, the technology investments and support from IT staff will keep operations smooth. With this foundation, the facility can effectively support both educational and administrative needs, providing a secure, reliable environment for students, faculty, and visitors alike.

3.3 Technical Feasibility

Based on the information that has been gathered, the project appears to be technically feasible. The network requirements, hardware and software including devices for specialised labs (Cisco device and Embedded lab devices) are well known. The monitoring, management, and reporting tools that are needed for network maintenance also has been identified, which we will use the SolarWinds Network Performance Monitor (NPM). To ensure that the coverage of wireless networks is able to support the area of the faculty is by identifying the location of most congested access points and areas with poor signal coverage. Next, The positioning of the network device to optimise the network performance also has been mentioned where it will install the Wi-Fi access points that connect to nearby switches. All the devices that are needed for all labs, conferencing room, and hybrid classrooms has been identified clearly and how to maintain it also has been identified. Therefore, in terms of technical, it is well prepared to conduct the project.

4.0 Meeting Minute

MEETING MINUTE 1

DATE/TIME		6th november 2024	
LOCATION		ONLINE - GMEET	
AGENDA		1) Discuss the question given for Task 2 2) Divide question for each member	
Meeting MC		YASMIN BATRISYIA BINTI ZAHIRUDDIN	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI		2:00 PM	-
AHMAD ADIB ZIKRI BIN A.MAZLAM		2:00 PM	-
YASMIN BATRISYIA BINTI ZAHIRUDDIN		2:00 PM	-
SYARIFAH DANIA BINTI SYED ABU BAKAR		2:00 PM	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discuss the materials given for Task 2	i) Syarifah Dania suggested all of the team members go through the questions and determine the requirements needed to complete Task 1. ii) All members read the materials given and identify the requirements	All members (6/11)
2	Provide question and answer	i) Yasmin Batrisyia and Syarifah Dania suggest each member to search for questions and answer.	Yasmin Batrisyia (6/11)

		ii) All members find question and answer to complete the task	Muhammad Mukhritz Al Iman (6/11)
3	Next meeting	i) All members agreed to conduct another meeting on 2nd of November as everybody is currently busy with programs ii) Agenda of the next meeting was decided. iii) Syarifah Dania reminded all members to do some research about the project to ensure that everyone will have more understanding before next meeting	All members (6/11)
5	Meeting ended	After completing the dividing task between each member, Ahmad Adib ended the meeting.	Ahmad Adib (31/10)

MEETING MINUTE 2

DATE/TIME		8nd of November 2024	
LOCATION		Ar-Rayyan Cafe, KTDI	
AGENDA		1) Analyse question from each member 2) Answer the question 3) Make a feasibility 4) Finalise the report.	
Meeting MC		YASMIN BATRISYIA BINTI ZAHIRUDDIN	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI		9:00 PM	-
AHMAD ADIB ZIKRI BIN A.MAZLAM		9:00 PM	-
YASMIN BATRISYIA BINTI ZAHIRUDDIN		9:00 PM	-
SYARIFAH DANIA BINTI SYED ABU BAKAR		9:00 PM	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Analyse question prepared by each member	i) Muhammad Mukhritz prepared list of all question that has been prepared by each member ii) All members discuss and choose which question that is suitable to be put in the report	Muhammad Mukhritz Al Iman (8/11)
2	Task Distribution for each members	i) Muhammad Mukhritz prepared a poll for all members to choose which part that they want to do.	Syarifah Dania (8/11)

		ii) Each member chose what feasibility that want to do	Muhammad Mukhriz Al Iman (8/11)
3	Reporting	i) Task distribution for the report was done ii) Cover page, table of content and Introduction were done by Muhammad Mukhriz. iii) Meeting minutes was done by Syarifah Dania.	All members (8/11)
6	Meeting ended	Yasmin Batrisyia ended the meeting once the report was completed.	Yasmin Batrisyia (8/11)